

ABHINAV HARBOLA

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PROFESSIONAL SUMMARY

ML Engineer building production-grade ML and LLM systems across recommender systems, retrieval-augmented generation, autonomous agents, and MLOps governance. Delivered a self-governing pipeline with automated drift detection and statistical model gating, a guardrailed RAG system with fail-closed safety gates and multi-provider failover, and a multi-agent research system with fully sourced, traceable outputs. Experienced end-to-end training through deployment and monitoring, on free-tier infrastructure. Proficient in Python and its ML/LLM ecosystem.

TECHNICAL SKILLS

Languages: Python, SQL, C++

Machine Learning & Deep Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Hugging Face Transformers, Two-Tower & Sequential (SASRec) Recommenders, SHAP, LIME, Statistical Significance Testing (McNemar, Bootstrap CI)

Generative AI & LLM Systems: LangGraph, RAG Pipelines, Multi-Agent Orchestration, Prompt Engineering, Few-Shot Prompting, Google Gemini API, Groq, NVIDIA NIM, FAISS, Qdrant, Vector Embeddings, RAGAS

MLOps & Production Systems: Automated Drift Detection (Evidently, KS-test), MLflow, DVC, Model Monitoring, CI/CD, Docker, Model Governance & Rollback, Inference Optimization, LLMOps

Data, APIs & Infrastructure: FastAPI, Streamlit, PostgreSQL (Neon), SQLite, Pandas, NumPy, Matplotlib, Plotly, Tableau, REST APIs, AWS, Explainable AI (XAI)

PROJECTS

Continuous Credit Risk Governance Pipeline | *Python, Postgres, MLflow, Evidently, DVC, FastAPI, Streamlit* 2026

- Engineered a self-governing MLOps pipeline that simulates the full credit-risk model lifecycle, drift detection, retraining, statistical gating, promotion, and rollback, on 150K+ Kaggle credit records streamed across 637 live batches with zero human intervention.
- Implemented Evidently-based drift detection with an empirically tuned significance threshold that cut false-retrain triggers from ~40% to ~1% of undrifted batches while reliably catching real injected drift.
- Built a three-gate model promotion system (tolerance band, dominance, McNemar/bootstrap significance testing) and a bootstrap-CI-confirmed rollback mechanism, logging every governance decision to a Postgres audit trail validated by 55 automated tests.

Kubernetes Gated RAG | *Python, LangGraph, Qdrant, FlashRank, Groq, Gemini API, Streamlit* 2025

- Built a defense-in-depth RAG system for Kubernetes Q&A as a LangGraph state machine with fail-closed safety and topic gates, threshold-gated cross-encoder reranking, and a 3-provider LLM failover chain, running entirely on a no-GPU, 16GB machine.
- Designed a two-layer exact + semantic caching system over Qdrant using 768-dimensional Gemini embeddings and a cosine ≥ 0.95 semantic-match threshold, eliminating redundant retrieval and generation for repeated and paraphrased queries.
- Evaluated retrieval and generation quality with a RAGAS-based harness across 6 metrics (faithfulness, answer relevancy, context precision/recall) judged by an isolated model family, with full node- and provider-level tracing via Logfire.

Competitive Intelligence Agent (CIA) | *Python, LangGraph-style Multi-Agent, NIM, Groq, Gemini, Postgres* 2025

- Designed an autonomous Planner-Executor-Critic-Synthesizer research agent that produces fully sourced competitive intelligence briefs across 5 required fields, self-correcting via a bounded critique/replan loop (max 3 cycles) with hard stops on tool calls and wall-clock time.
- Orchestrated 4 distinct LLM model families across 3 independent providers (NIM, Groq, Gemini) to keep the evaluation judge structurally isolated from the components it grades, backed by a Postgres memory layer for cache-aware reuse across runs.

- Built an evaluation harness with an isolated LLM judge scoring groundedness and completeness across a 15-company benchmark, running ablation studies that measured the Critic loop's quantified impact on output quality.

ReelBench: Multi-Stage Recommender Benchmark | *Python, PyTorch, FAISS, LightGBM, FastAPI, Streamlit* 2025

- Built a production-style two-stage recommender system on MovieLens 25M, benchmarking 5 approaches (popularity, item-item CF, ALS/BPR, two-tower neural retriever, SASRec) head-to-head under one leakage-tested evaluation harness (Recall/NDCG/MAP@10/20, coverage, diversity).
- Engineered a RAM-safe item-item collaborative filtering model using row-blocked sparse similarity with bounded top-K, cutting memory use from a projected ~15GB dense matrix to under 200MB at scale.
- Served a FAISS-based two-tower retrieval + LightGBM re-ranking pipeline via FastAPI on CPU-only hardware, plus a separate Gemini-embedding cold-start index enabling recommendations for zero-interaction-history items.

EXPERIENCE

Data Science Intern | Coding Blocks Jun 2024 - Aug 2024

- Developed an XGBoost churn prediction model to identify at-risk customers, improving baseline accuracy by 9% through GridSearchCV hyperparameter tuning and feature engineering on an imbalanced dataset.
- Automated extraction of 5,000+ competitor reviews using Python web scraping, converting raw HTML into structured Pandas DataFrames to support downstream sentiment modeling.

EDUCATION

B.Tech in Computer Science and Engineering Aug 2022 - May 2026 (Expected)

Lovely Professional University, Punjab, India | CGPA: 8.19 / 10.0

CERTIFICATIONS AND ACHIEVEMENTS

- IBM DevOps & Software Engineering Professional Certificate - Coursera, Nov 2025
- NPTEL Cloud Computing - IIT Kharagpur, Oct 2024
- Google IT Automation with Python - Coursera, Dec 2023
- Dean's List - Top 10% of students for 3 consecutive years (2023, 2024, 2025)
- 5-Star Badge in Python and C++ - HackerRank